LVIV POLYTECHNIC NATIONAL UNIVERSITY

Computer Science

Exploring the foundation of modern computing architecture and software engineering.

Herasymenko Vadym KNIK-11

THE CORE FOUNDATION

Beyond Just Writing Code



Algorithmic Logic

Mastering complex data structures.



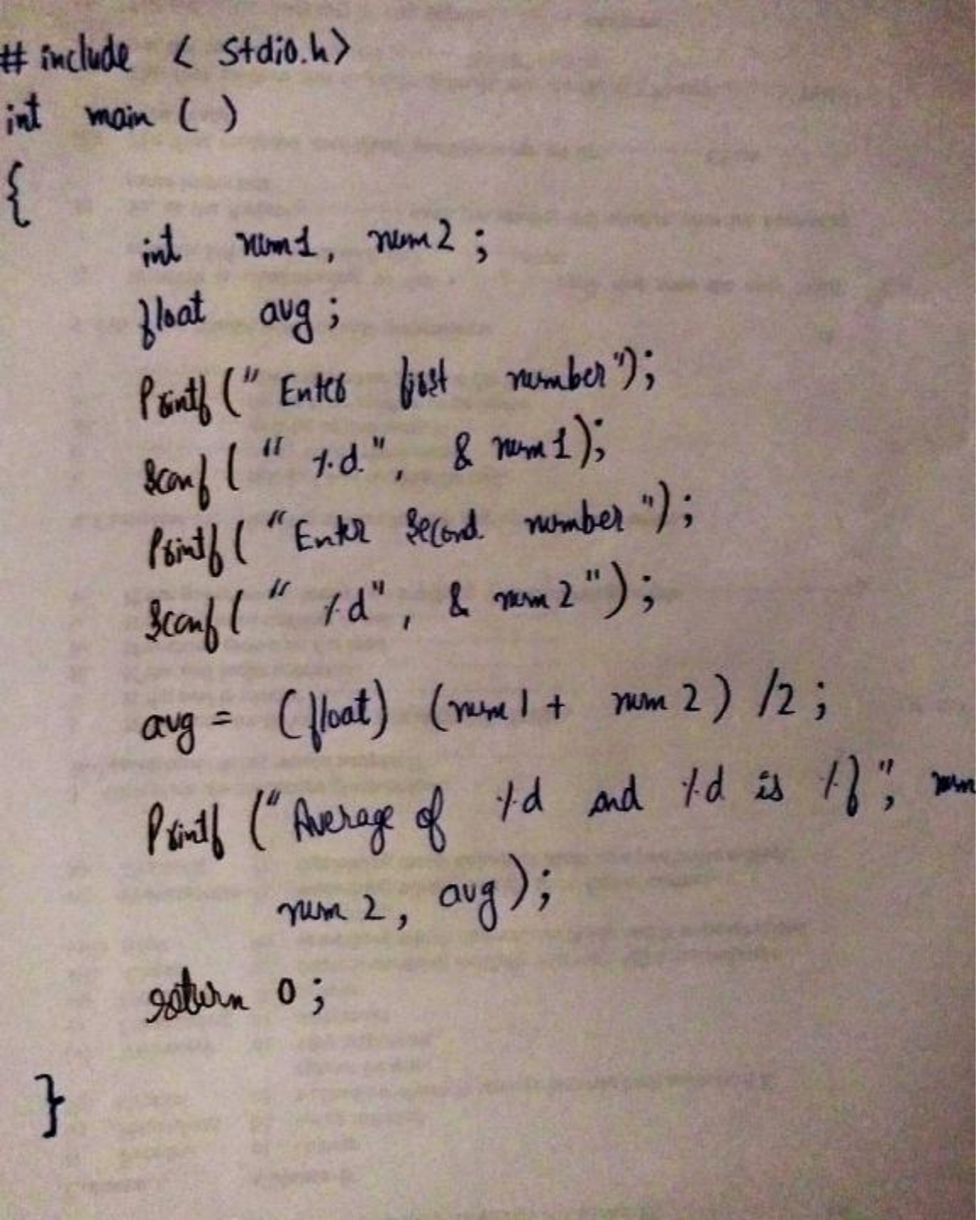
System Architecture

Understanding hardware-software bridges.



Scalable Engineering

Building robust, real-world applications.



Programming on Paper

- **Mental Compilation: Executing logic without an IDE.**
- **Pure Focus: Writing flawless syntax under exam pressure.**

> *"Huge thanks to Zoreslava Yaroslavivna Shpak for this unforgettable, foundational experience."*

Bringing Code to Life

- Embedded C/C++: Programming physical ESP32 microcontrollers.
- Internet of Things: Connecting sensors to the web.
- Instant Feedback: Controlling the physical world directly.



```
MONITOR FOR 6802 1.4          9-14-80  TSC ASSEMBLER  PAGE    2


C000                    ORG     ROM+$0000 BEGIN MONITOR
C000 8E 00 70  START    LDS     #STACK

               ************************************
               * FUNCTION: INITA - Initialize ACIA
               * INPUT: none
               * OUTPUT: none
               * CALLS: none
               * DESTROYS: acc A

0013           RESETA   EQU     %00010011
0011           CTLREG   EQU     %00010001

C003 86 13     INITA    LDA A   #RESETA    RESET ACIA
C005 B7 80 04           STA A   ACIA
C008 86 11              LDA A   #CTLREG    SET 8 BITS AND 2 STOP
C00A B7 80 04           STA A   ACIA

C00D 7E C0 F1           JMP     SIGNON     GO TO START OF MONITOR

               ************************************
               * FUNCTION: INCH - Input character
               * INPUT: none
               * OUTPUT: char in acc A
               * DESTROYS: acc A
               * CALLS: none
               * DESCRIPTION: Gets 1 character from terminal

C010 B6 80 04  INCH     LDA A   ACIA       GET STATUS
C013 47                 ASR A              SHIFT RDRF FLAG INTO CARRY
C014 24 FA              BCC     INCH       RECIEVE NOT READY
C016 B6 80 05           LDA A   ACIA+1     GET CHAR
C019 84 7F              AND A   #$7F       MASK PARITY
C01B 7E C0 79           JMP     OUTCH      ECHO & RTS

               ************************************
               * FUNCTION: INHEX - INPUT HEX DIGIT
               * INPUT: none
               * OUTPUT: Digit in acc A
               * CALLS: INCH
               * DESTROYS: acc A
               * Returns to monitor if not HEX input

C01E 8D F0     INHEX    BSR     INCH       GET A CHAR
C020 81 30              CMP A   #'0        ZERO
C022 2B 11              BMI     HEXERR     NOT HEX
C024 81 39              CMP A   #'9        NINE
C026 2F 0A              BLE     HEXRTS     GOOD HEX
C028 81 41              CMP A   #'A
C02A 2B 09              BMI     HEXERR     NOT HEX
C02C 81 46              CMP A   #'F
C02E 2E 05              BGT     HEXERR
C030 80 07              SUB A   #7         FIX A-F
C032 84 0F     HEXRTS   AND A   #$0F       CONVERT ASCII TO DIGIT
C034 39                 RTS

C035 7E C0 AF  HEXERR   JMP     CTRL       RETURN TO CONTROL LOOP
```

Learning Assembly

```
MOV AX, 0x0001
ADD AX, BX
; Unforgettable emotions
```

- **Low-Level Access: Direct CPU register manipulation.**
- **Bypassing Abstractions: Understanding how memory truly works.**

Where Does This Lead?



Software Engineer

Building high-level applications.



Embedded Developer

Programming IoT and hardware.



Systems Architect

Designing complex IT infrastructures.



Thank You!